# TabuLLM: AI Benchmark Report

## 1. Executive Summary: Consolidated Audit Results

This report consolidates the results of the TabuLLM data augmentation pipeline. The objective is to audit the fidelity, diversity, and privacy of the synthetic data generated via LLM inference. Below are the averaged performance indicators across all processed datasets in this session.

| Audit Indicator | Global Session Mean (± Std) |
|---|---|
| Synthetic Fidelity (JSD) | 63.0 ± 3.61 |
| Correlation Stability | 88.3 ± 0.90 |
| Distribution Consistency | 75.0 ± 3.55 |
| Entropy Preservation | 67.4 ± 1.43 |
| Mean Z Score Drift | 0.5 ± 0.03 |
| Categorical Fidelity | 0.0 ± 0.00 |
| Privacy Risk | 5.7 ± 2.41 |
| Data Density | 100.0 ± 0.00 |
| Coverage Score | 88.2 ± 2.93 |
| Feature Accuracy | 53.7 ± 2.00 |
| Novelty Score | 0.7 ± 0.03 |

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## Dataset Audit: heart_failure_clinical_records.csv

### 1. Dataset Portrait

| Attribute | Value |
|---|---|
| Rows | 299 |
| Columns | 13 |
| Numeric features | 13 |
| Categorical features | 0 |
| Missing cells (%) | 0.0% |

### 2. Generation Strategy & LLM Config [1]

| Parameter | Configuration |
|---|---|
| Algorithm (*12) | google/gemma-4-e2b |
| Folds (*13) | 10 |
| Context % (*14) | 0% |
| Generation % (*15) | 100% |
| Join context (*16) | False |

### 3. Statistical Significance & Data Drift

| Feature | Orig Mean | Aug Mean | Shift | P-Value | Status |
|---|---|---|---|---|---|
| age | 60.834 | 65.236 | +4.402 | 7.2% | OK |
| anaemia | 0.431 | 0.232 | -0.199 | 46.2% | Drift |
| creatinine_phosphokinase | 581.839 | 713.616 | +131.777 | 22.6% | Warn |
| diabetes | 0.418 | 0.152 | -0.267 | 63.8% | Drift |
| ejection_fraction | 38.084 | 33.152 | -4.932 | 13.0% | Warn |
| high_blood_pressure | 0.351 | 0.202 | -0.149 | 42.5% | Drift |
| platelets | 263358.029 | 248392.630 | -14965.400 | 5.7% | OK |
| serum_creatinine | 1.394 | 1.866 | +0.473 | 33.9% | Drift |
| serum_sodium | 136.625 | 134.646 | -1.979 | 1.4% | OK |
| sex | 0.649 | 0.384 | -0.265 | 40.8% | Drift |
| smoking | 0.321 | 0.020 | -0.301 | 93.7% | Drift |
| time | 130.261 | 67.394 | -62.867 | 48.3% | Drift |

### 4. Quality & Fidelity Metrics (Mean ± Std)

| Metric Category | Aggregated Score |
|---|---|
| Synthetic Fidelity (JSD) (*1) | 63.0 ± 3.61 |
| Correlation Stability (*2) | 88.3 ± 0.90 |
| Distribution Consistency (*3) | 75.0 ± 3.55 |
| Entropy Preservation (*4) | 67.4 ± 1.43 |
| Mean Z Score Drift (*5) | 0.5 ± 0.03 |
| Categorical Fidelity (*6) | 0.0 ± 0.00 |
| Privacy Risk (*7) | 5.7 ± 2.41 |

| | |
|---|---|
| Data Density (*8) | 100.0 ± 0.00 |
| Coverage Score (*9) | 88.2 ± 2.93 |
| Feature Accuracy (*10) | 53.7 ± 2.00 |
| Novelty Score (*11) | 0.7 ± 0.03 |

*[1] (*12) Algorithm: Contextual constraint or model parameter.*

*[2] (*13) Folds: Contextual constraint or model parameter.*

*[3] (*14) Context %: Contextual constraint or model parameter.*

*[4] (*15) Generation %: Contextual constraint or model parameter.*

*[5] (*16) Join context: Contextual constraint or model parameter.*

*[6] Synthetic Fidelity (JSD): Overall distribution similarity.*
*[7] Correlation Stability: Multivariate relationships preserved.*
*[8] Distribution Consistency: Median feature similarity.*
*[9] Entropy Preservation: Information diversity maintained.*
*[10] Mean Z Score Drift: Standardized distribution shift.*
*[11] Categorical Fidelity: Category distribution similarity.*
*[12] Privacy Risk: Synthetic rows too close to originals.*
*[13] Data Density: Non-missing cells in augmented data.*
*[14] Coverage Score: Synthetic coverage of original space.*
*[15] Feature Accuracy: Mean feature similarity.*
*[16] Novelty Score: Distance ratio to nearest neighbors.*

## 6. Downstream ML Utility: Comparative Analysis

### Model: LR

| Metric | Real (Baseline) | Generated | Augmented | Delta (Lift) | Sig. |
|---|---|---|---|---|---|
| ACCURACY | 0.8066666666666666 | N/A | 0.7783333333333333 | -0.02833333333333332 | |
| F1 | 0.6937863401621888 | N/A | 0.7086714364937362 | 0.014885096331547365 | |

### Model: DT

| Metric | Real (Baseline) | Generated | Augmented | Delta (Lift) | Sig. |
|---|---|---|---|---|---|
| ACCURACY | 0.79 | N/A | 0.7683333333333333 | -0.021666666666666667 | |
| F1 | 0.681956629831503 | N/A | 0.6696355563913111 | -0.012321073440191876 | |

### Model: XGB

| Metric | Real (Baseline) | Generated | Augmented | Delta (Lift) | Sig. |
|---|---|---|---|---|---|
| ACCURACY | 0.835 | N/A | 0.8433333333333334 | 0.008333333333333337 | |
| F1 | 0.7547418227770806 | N/A | 0.7773865902636359 | 0.022644767486555523 | |

### Model: GB

| Metric | Real (Baseline) | Generated | Augmented | Delta (Lift) | Sig. |
|---|---|---|---|---|---|
| ACCURACY | 0.8300000000000001 | N/A | 0.8316666666666667 | 0.001666666666666607 | |
| F1 | 0.7541632435801531 | N/A | 0.7648572827382548 | 0.010694039158101664 | |

### Model: AB

| Metric | Real (Baseline) | Generated | Augmented | Delta (Lift) | Sig. |
|---|---|---|---|---|---|
| ACCURACY | 0.825 | N/A | 0.8266666666666668 | 0.001666666666666607 | |
| F1 | 0.7417404771442427 | N/A | 0.7535785909067643 | 0.011838113762521552 | |

### Model: KNN

| Metric | Real (Baseline) | Generated | Augmented | Delta (Lift) | Sig. |
|---|---|---|---|---|---|
| ACCURACY | 0.6133333333333335 | N/A | 0.5349999999999999 | -0.07833333333333332 | * |
| F1 | 0.007142857142857143 | N/A | 0.4356021459961791 | 0.42845928885332196 | * |

### Model: NB

| Metric | Real (Baseline) | Generated | Augmented | Delta (Lift) | Sig. |
|---|---|---|---|---|---|
| ACCURACY | 0.7499999999999999 | N/A | 0.7849999999999999 | 0.03499999999999996 | |
| F1 | 0.5667171896628354 | N/A | 0.7052746879637636 | 0.13855749830092828 | * |

### Model: MLP

| Metric | Real (Baseline) | Generated | Augmented | Delta (Lift) | Sig. |
|---|---|---|---|---|---|
| ACCURACY | 0.5116666666666666 | N/A | 0.4683333333333327 | -0.04333333333333332 | |
| F1 | 0.2046209604349139 | N/A | 0.3600637359437734 | 0.1554427755088595 | |

### Model: SVM

| Metric | Real (Baseline) | Generated | Augmented | Delta (Lift) | Sig. |
|---|---|---|---|---|---|
| ACCURACY | 0.6333333333333334 | N/A | 0.6066666666666667 | -0.02666666666666665 | |
| F1 | N/A | N/A | 0.2726638357060673 | 0.27266383570606173 | * |

### Model: RF

| Metric | Real (Baseline) | Generated | Augmented | Delta (Lift) | Sig. |
|---|---|---|---|---|---|
| ACCURACY | 0.8316666666666667 | N/A | 0.8450000000000001 | 0.01333333333333333 | |
| F1 | 0.7335384082180338 | N/A | 0.7814474922753252 | 0.0479090840572912 | |

## Appendix: Benchmark Data Trace

*Legend: [Blue] = Real Context | [Green] = Generated Synthetic | [Yellow] = Real Training (Remaining) | [Grey] = Real Test Baseline*

### A. Baseline: Original Dataset (Full Record Sample)

| # | age | anaemia | creatinine_phosphoki | diabetes | ejection_fractio | high_blood_pressu | platelets | serum_creatinine | serum_sodium | sex |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 75.0 | 0.0 | 582.0 | 0.0 | 20.0 | 1.0 | 265000.0 | 1.9 | 130.0 | 1.0 |
| 2 | 55.0 | 0.0 | 7861.0 | 0.0 | 38.0 | 0.0 | 263358.03 | 1.1 | 136.0 | 1.0 |
| 3 | 65.0 | 0.0 | 146.0 | 0.0 | 20.0 | 0.0 | 162000.0 | 1.3 | 129.0 | 1.0 |
| 4 | 50.0 | 1.0 | 111.0 | 0.0 | 20.0 | 0.0 | 210000.0 | 1.9 | 137.0 | 1.0 |
| 5 | 65.0 | 1.0 | 160.0 | 1.0 | 20.0 | 0.0 | 327000.0 | 2.7 | 116.0 | 0.0 |
| 6 | 90.0 | 1.0 | 47.0 | 0.0 | 40.0 | 1.0 | 204000.0 | 2.1 | 132.0 | 1.0 |
| 7 | 75.0 | 1.0 | 246.0 | 0.0 | 15.0 | 0.0 | 127000.0 | 1.2 | 137.0 | 1.0 |
| 8 | 60.0 | 1.0 | 315.0 | 1.0 | 60.0 | 0.0 | 454000.0 | 1.1 | 131.0 | 1.0 |
| 9 | 65.0 | 0.0 | 157.0 | 0.0 | 65.0 | 0.0 | 263358.03 | 1.5 | 138.0 | 0.0 |
| 10 | 80.0 | 1.0 | 123.0 | 0.0 | 35.0 | 1.0 | 388000.0 | 9.4 | 133.0 | 1.0 |
| 11 | 75.0 | 1.0 | 81.0 | 0.0 | 38.0 | 1.0 | 368000.0 | 4.0 | 131.0 | 1.0 |
| 12 | 62.0 | 0.0 | 231.0 | 0.0 | 25.0 | 1.0 | 253000.0 | 0.9 | 140.0 | 1.0 |
| 13 | 45.0 | 1.0 | 981.0 | 0.0 | 30.0 | 0.0 | 136000.0 | 1.1 | 137.0 | 1.0 |
| 14 | 50.0 | 1.0 | 168.0 | 0.0 | 38.0 | 1.0 | 276000.0 | 1.1 | 137.0 | 1.0 |
| 15 | 49.0 | 1.0 | 80.0 | 0.0 | 30.0 | 1.0 | 427000.0 | 1.0 | 138.0 | 0.0 |
| 16 | 82.0 | 1.0 | 379.0 | 0.0 | 50.0 | 0.0 | 47000.0 | 1.3 | 136.0 | 1.0 |
| 17 | 87.0 | 1.0 | 149.0 | 0.0 | 38.0 | 0.0 | 262000.0 | 0.9 | 140.0 | 1.0 |
| 18 | 45.0 | 0.0 | 582.0 | 0.0 | 14.0 | 0.0 | 166000.0 | 0.8 | 127.0 | 1.0 |
| 19 | 70.0 | 1.0 | 125.0 | 0.0 | 25.0 | 1.0 | 237000.0 | 1.0 | 140.0 | 0.0 |
| 20 | 48.0 | 1.0 | 582.0 | 1.0 | 55.0 | 0.0 | 87000.0 | 1.9 | 121.0 | 0.0 |

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## B.1 Audit Trace: Fold 1 | Gen: 100%

| # | age | anaemia | creatinine_phosphoki | diabetes | ejection_fractio | high_blood_pressu | platelets | serum_creatinine | serum_sodium | sex |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 46.2753111277481 9 | 0.0 | 1882.0 | 1.0 | 31.0 | 0.0 | 330174.066616755 44 | 1.07449377744503 63 | 136.0 | 0.0 |
| 2 | 49.5667776745159 7 | 0.0 | 482.0 | 0.0 | 28.0 | 1.0 | 319000.0 | 1.04332223254840 33 | 131.0 | 0.0 |
| 3 | 75.1069286375004 2 | 1.0 | 578.0 | 0.0 | 30.0 | 0.0 | 225007.128575833 37 | 1.82408328205831 06 | 134.0 | 1.0 |
| 4 | 69.4401457083421 9 | 1.0 | 316.0 | 0.0 | 36.0 | 0.0 | 222160.016189815 8 | 1.46400647592631 96 | 141.0 | 1.0 |
| 5 | 73.5271481456526 6 | 0.0 | 914.0 | 1.0 | 33.0 | 0.0 | 263358.03 | 2.31800491440708 62 | 135.0 | 0.0 |
| 6 | 60.6997714115125 86 | 0.0 | 1773.0 | 0.0 | 27.0 | 0.0 | 364020.320023882 4 | 2.04401828707899 34 | 143.0 | 0.0 |
| 7 | 45.0 | 1.0 | 66.0 | 1.0 | 25.0 | 0.0 | 233000.0 | 0.8 | 135.0 | 1.0 |
| 8 | 70.0 | 0.0 | 161.0 | 0.0 | 25.0 | 0.0 | 244000.0 | 1.2 | 142.0 | 0.0 |
| 9 | 58.0 | 1.0 | 400.0 | 0.0 | 40.0 | 0.0 | 164000.0 | 1.0 | 139.0 | 0.0 |
| 10 | 55.0 | 0.0 | 572.0 | 1.0 | 35.0 | 0.0 | 231000.0 | 0.8 | 143.0 | 0.0 |

## B.2 Audit Trace: Fold 2 | Gen: 100%

| # | age | anaemia | creatinine_phosphoki | diabetes | ejection_fractio | high_blood_pressu | platelets | serum_creatinine | serum_sodium | sex |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 77.3321507887709 7 | 0.0 | 582.0 | 0.0 | 31.0 | 0.0 | 263358.03 | 1.83 | 134.0 | 0.0 |
| 2 | 79.6435723598324 5 | 0.0 | 153.0 | 0.0 | 38.0 | 0.0 | 149047.523685355 67 | 1.88455480225940 59 | 143.0 | 0.0 |
| 3 | 57.6438696554128 | 0.0 | 512.0 | 0.0 | 30.0 | 0.0 | 99003.6296546591 6 | 1.49748213792908 55 | 135.0 | 1.0 |
| 4 | 49.8777572803329 2 | 0.0 | 607.0 | 0.0 | 35.0 | 0.0 | 311100.184477003 7 | 1.80220582426633 81 | 135.0 | 1.0 |
| 5 | 45.9421482315606 5 | 0.0 | 582.0 | 0.0 | 20.0 | 0.0 | 163550.414597942 3 | 0.76231407073757 38 | 128.0 | 0.0 |
| 6 | 64.0133637026342 2 | 0.0 | 380.0 | 0.0 | 25.0 | 0.0 | 265723.533284734 9 | 1.19342242468422 82 | 136.0 | 1.0 |
| 7 | 60.0 | 0.0 | 2656.0 | 1.0 | 30.0 | 0.0 | 305000.0 | 2.3 | 137.0 | 1.0 |
| 8 | 45.0 | 1.0 | 130.0 | 0.0 | 35.0 | 0.0 | 174000.0 | 0.8 | 139.0 | 1.0 |
| 9 | 45.0 | 0.0 | 582.0 | 1.0 | 55.0 | 0.0 | 543000.0 | 1.0 | 132.0 | 0.0 |
| 10 | 65.0 | 1.0 | 59.0 | 1.0 | 60.0 | 0.0 | 172000.0 | 0.9 | 137.0 | 0.0 |

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## B.3 Audit Trace: Fold 3 | Gen: 100%

| # | age | anaemia | creatinine_phosphoki | diabetes | ejection_fractio | high_blood_pressu | platelets | serum_creatinine | serum_sodium | sex |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 70.7772022542234 | 0.0 | 599.0 | 0.0 | 21.0 | 0.0 | 263358.03 | 1.77326292354416 9 | 134.0 | 1.0 |
| 2 | 55.9969922410112 06 | 0.0 | 822.0 | 1.0 | 25.0 | 0.0 | 71668.3376438826 7 | 1.89949870683520 14 | 136.0 | 0.0 |
| 3 | 63.4089784039375 4 | 0.0 | 578.0 | 0.0 | 31.0 | 0.0 | 162684.262180621 56 | 0.89797729786527 87 | 136.0 | 0.0 |
| 4 | 53.4921024829728 9 | 0.0 | 1708.0 | 0.0 | 23.0 | 0.0 | 315256.581264189 27 | 1.1 | 137.0 | 1.0 |
| 5 | 58.1117440447324 26 | 0.0 | 61.0 | 0.0 | 37.0 | 0.0 | 153143.670914655 96 | 5.72816454267201 3 | 133.0 | 1.0 |
| 6 | 54.2183167379814 75 | 1.0 | 112.0 | 0.0 | 18.0 | 0.0 | 206890.841631009 27 | 0.89272277540061 75 | 138.0 | 0.0 |
| 7 | 50.0 | 0.0 | 582.0 | 1.0 | 38.0 | 0.0 | 310000.0 | 1.9 | 135.0 | 1.0 |
| 8 | 79.0 | 1.0 | 55.0 | 0.0 | 50.0 | 1.0 | 172000.0 | 1.8 | 133.0 | 1.0 |
| 9 | 65.0 | 1.0 | 160.0 | 1.0 | 20.0 | 0.0 | 327000.0 | 2.7 | 116.0 | 0.0 |
| 10 | 70.0 | 0.0 | 92.0 | 0.0 | 60.0 | 1.0 | 317000.0 | 0.8 | 140.0 | 0.0 |

## B.4 Audit Trace: Fold 4 | Gen: 100%

| # | age | anaemia | creatinine_phosphoki | diabetes | ejection_fractio | high_blood_pressu | platelets | serum_creatinine | serum_sodium | sex |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 58.4258421249398 05 | 0.0 | 528.0 | 0.0 | 50.0 | 0.0 | 125089.397961136 9 | 0.78431529648305 82 | 134.0 | 0.0 |
| 2 | 70.0435044828257 8 | 0.0 | 127.0 | 0.0 | 25.0 | 1.0 | 236956.495517174 22 | 1.03262836211933 17 | 139.0 | 0.0 |
| 3 | 68.8767012003207 8 | 0.0 | 463.0 | 1.0 | 46.0 | 1.0 | 185673.979279807 54 | 1.15506804801283 12 | 139.0 | 1.0 |
| 4 | 45.0 | 0.0 | 7198.0 | 0.0 | 25.0 | 0.0 | 389646.310300509 9 | 1.0 | 139.0 | 1.0 |
| 5 | 69.2183037777744 8 | 0.0 | 547.0 | 0.0 | 20.0 | 0.0 | 266582.143407398 6 | 1.18544641481503 43 | 134.0 | 1.0 |
| 6 | 79.4472335019760 9 | 0.0 | 792.0 | 0.0 | 37.0 | 0.0 | 263358.03 | 1.14035195435574 59 | 134.0 | 1.0 |
| 7 | 48.0 | 1.0 | 131.0 | 1.0 | 30.0 | 1.0 | 244000.0 | 1.6 | 130.0 | 0.0 |
| 8 | 95.0 | 1.0 | 371.0 | 0.0 | 30.0 | 0.0 | 461000.0 | 2.0 | 132.0 | 1.0 |
| 9 | 45.0 | 1.0 | 981.0 | 0.0 | 30.0 | 0.0 | 136000.0 | 1.1 | 137.0 | 1.0 |
| 10 | 50.0 | 0.0 | 582.0 | 1.0 | 38.0 | 0.0 | 310000.0 | 1.9 | 135.0 | 1.0 |

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## B.5 Audit Trace: Fold 5 | Gen: 100%

| # | age | anaemia | creatinine_phosphoki | diabetes | ejection_fractio | high_blood_pressu | platelets | serum_creatinine | serum_sodium | sex |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 64.6848063402939 | 1.0 | 246.0 | 1.0 | 25.0 | 0.0 | 197873.92253611758 | 1.4693426051353398 | 129.0 | 0.0 |
| 2 | 58.05801240051275 | 0.0 | 186.0 | 0.0 | 21.0 | 0.0 | 121200.91927941889 | 2.6669614880615304 | 128.0 | 1.0 |
| 3 | 82.53993661248599 | 1.0 | 88.0 | 0.0 | 32.0 | 0.0 | 198848.01448857464 | 0.8931994566784512 | 138.0 | 0.0 |
| 4 | 51.41712473478663 | 0.0 | 287.0 | 0.0 | 37.0 | 0.0 | 216531.421775545 | 2.1051453489668384 | 132.0 | 0.0 |
| 5 | 73.85467957999738 | 0.0 | 301.0 | 0.0 | 45.0 | 0.0 | 230548.769008006 3 | 2.2403448588003 67 | 138.0 | 0.0 |
| 6 | 84.33040713684436 | 0.0 | 28.0 | 0.0 | 45.0 | 0.0 | 359598.24428210664 | 2.9241128088423602 | 132.0 | 0.0 |
| 7 | 78.0 | 0.0 | 224.0 | 0.0 | 50.0 | 0.0 | 481000.0 | 1.4 | 138.0 | 1.0 |
| 8 | 70.0 | 0.0 | 582.0 | 0.0 | 20.0 | 1.0 | 263358.03 | 1.83 | 134.0 | 1.0 |
| 9 | 61.0 | 0.0 | 248.0 | 0.0 | 30.0 | 1.0 | 267000.0 | 0.7 | 136.0 | 1.0 |
| 10 | 40.0 | 0.0 | 478.0 | 1.0 | 30.0 | 0.0 | 303000.0 | 0.9 | 136.0 | 1.0 |

## B.6 Audit Trace: Fold 6 | Gen: 100%

| # | age | anaemia | creatinine_phosphoki | diabetes | ejection_fractio | high_blood_pressu | platelets | serum_creatinine | serum_sodium | sex |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 74.95877675233974 | 0.0 | 582.0 | 0.0 | 29.0 | 1.0 | 263358.03 | 1.83 | 134.0 | 0.0 |
| 2 | 56.42945707279832 | 1.0 | 204.0 | 0.0 | 50.0 | 0.0 | 152214.72853639917 | 7.05643434176134 | 135.0 | 0.0 |
| 3 | 60.71332722595645 5 | 0.0 | 515.0 | 1.0 | 36.0 | 0.0 | 457562.6104787994 | 0.7754784975852872 | 137.0 | 1.0 |
| 4 | 50.04555074701113 4 | 1.0 | 130.0 | 0.0 | 29.0 | 1.0 | 243349.14294413282 | 1.5442122523542419 | 130.0 | 0.0 |
| 5 | 47.12711218703093 | 1.0 | 954.0 | 0.0 | 29.0 | 0.0 | 136243.09853566068 | 1.3005562919200593 | 136.0 | 1.0 |
| 6 | 59.08357055825574 | 0.0 | 557.0 | 0.0 | 40.0 | 0.0 | 216908.35705582556 | 3.57169987815580 3 | 133.0 | 0.0 |
| 7 | 81.0 | 0.0 | 4540.0 | 0.0 | 35.0 | 0.0 | 231000.0 | 1.18 | 137.0 | 1.0 |
| 8 | 70.0 | 0.0 | 2695.0 | 1.0 | 40.0 | 0.0 | 241000.0 | 1.0 | 137.0 | 1.0 |
| 9 | 70.0 | 0.0 | 835.0 | 0.0 | 35.0 | 1.0 | 305000.0 | 0.8 | 133.0 | 0.0 |
| 10 | 80.0 | 0.0 | 776.0 | 1.0 | 38.0 | 1.0 | 192000.0 | 1.3 | 135.0 | 0.0 |

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## B.7 Audit Trace: Fold 7 | Gen: 100%

| # | age | anaemia | creatinine_phosphoki | diabetes | ejection_fractio | high_blood_pressu | platelets | serum_creatinine | serum_sodium | sex |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 68.2032323759734 | 0.0 | 1755.0 | 0.0 | 27.0 | 0.0 | 250302.942445228 | 0.92697057554771 | 137.0 | 1.0 |
| | 4 | | | | | | 92 | 07 | | |
| 2 | 71.1927370133137 | 0.0 | 582.0 | 0.0 | 20.0 | 1.0 | 263749.717678750 | 1.84669831818639 | 133.0 | 1.0 |
| | 3 | | | | | | 17 | 23 | | |
| 3 | 80.0469946703826 | 0.0 | 72.0 | 1.0 | 32.0 | 0.0 | 198621.408002623 | 1.17702346671038 | 132.0 | 1.0 |
| | 5 | | | | | | 04 | 4 | | |
| 4 | 68.4466859479450 | 0.0 | 556.0 | 0.0 | 20.0 | 0.0 | 265861.671486986 | 1.2 | 134.0 | 1.0 |
| | 8 | | | | | | 3 | | | |
| 5 | 63.6981769512490 | 0.0 | 222.0 | 1.0 | 52.0 | 0.0 | 189562.187658501 | 1.02603646097501 | 140.0 | 0.0 |
| | 9 | | | | | | 1 | 83 | | |
| 6 | 72.7761698813017 | 0.0 | 1219.0 | 0.0 | 26.0 | 0.0 | 243814.922007913 | 1.16298440158264 | 140.0 | 0.0 |
| | | | | | | | 2 | 4 | | |
| 7 | 75.0 | 0.0 | 582.0 | 1.0 | 30.0 | 1.0 | 263358.03 | 1.83 | 134.0 | 0.0 |
| 8 | 53.0 | 0.0 | 207.0 | 1.0 | 40.0 | 0.0 | 223000.0 | 1.2 | 130.0 | 0.0 |
| 9 | 60.0 | 1.0 | 2281.0 | 1.0 | 40.0 | 0.0 | 283000.0 | 1.0 | 141.0 | 0.0 |
| 10 | 59.0 | 1.0 | 280.0 | 1.0 | 25.0 | 1.0 | 302000.0 | 1.0 | 141.0 | 0.0 |

## B.8 Audit Trace: Fold 8 | Gen: 100%

| # | age | anaemia | creatinine_phosphoki | diabetes | ejection_fractio | high_blood_pressu | platelets | serum_creatinine | serum_sodium | sex |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 62.1245909437629 | 0.0 | 232.0 | 0.0 | 24.0 | 1.0 | 253012.459094376 | 0.90498363775051 | 139.0 | 1.0 |
| | 25 | | | | | | 3 | 7 | | |
| 2 | 69.7706164557680 | 1.0 | 135.0 | 1.0 | 33.0 | 0.0 | 222737.196202543 | 1.0 | 135.0 | 1.0 |
| | 8 | | | | | | 24 | | | |
| 3 | 82.2964666777076 | 0.0 | 1063.0 | 0.0 | 41.0 | 0.0 | 263358.03 | 2.53794463891511 | 135.0 | 0.0 |
| | 6 | | | | | | | 23 | | |
| 4 | 82.4106609955912 | 1.0 | 268.0 | 0.0 | 47.0 | 0.0 | 224248.614075905 | 1.46703624642515 | 140.0 | 1.0 |
| | 9 | | | | | | 67 | 14 | | |
| 5 | 54.7573723974709 | 0.0 | 1947.0 | 0.0 | 25.0 | 0.0 | 313805.897917976 | 1.13882041640464 | 137.0 | 1.0 |
| | 5 | | | | | | 8 | 78 | | |
| 6 | 68.7396829368334 | 0.0 | 571.0 | 1.0 | 34.0 | 0.0 | 227817.778055783 | 3.43492073420835 | 134.0 | 1.0 |
| | 3 | | | | | | 4 | 7 | | |
| 7 | 65.0 | 0.0 | 224.0 | 1.0 | 50.0 | 0.0 | 149000.0 | 1.3 | 137.0 | 1.0 |
| 8 | 80.0 | 0.0 | 898.0 | 0.0 | 25.0 | 0.0 | 149000.0 | 1.1 | 144.0 | 1.0 |
| 9 | 66.0 | 1.0 | 72.0 | 0.0 | 40.0 | 1.0 | 242000.0 | 1.2 | 134.0 | 1.0 |
| 10 | 75.0 | 1.0 | 582.0 | 0.0 | 30.0 | 0.0 | 225000.0 | 1.83 | 134.0 | 1.0 |

# TabuLLM: AI Benchmark Report

## B.9 Audit Trace: Fold 9 | Gen: 100%

| # | age | anaemia | creatinine_phosphoki | diabetes | ejection_fractio | high_blood_pressu | platelets | serum_creatinine | serum_sodium | sex |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 50.624824690134889 | 0.0 | 563.0 | 0.0 | 48.0 | 0.0 | 153374.89481408094 | 0.6291584855396283 | 133.0 | 0.0 |
| 2 | 60.187392970024307 | 0.0 | 79.0 | 0.0 | 19.0 | 0.0 | 74230.087455711882 | 2.310945527248172 | 134.0 | 1.0 |
| 3 | 56.698228861596377 | 0.0 | 2028.0 | 1.0 | 28.0 | 0.0 | 389253.320427548 6 | 1.373339786225709 | 136.0 | 1.0 |
| 4 | 69.428113249408601 | 0.0 | 123.0 | 0.0 | 44.0 | 1.0 | 283771.245299763 45 | 1.294281132494086 2 | 136.0 | 1.0 |
| 5 | 63.013211446383534 | 0.0 | 120.0 | 0.0 | 25.0 | 0.0 | 65986.788553616 45 | 2.208063800390352 3 | 134.0 | 1.0 |
| 6 | 54.290547120621284 | 0.0 | 302.0 | 0.0 | 43.0 | 0.0 | 153000.0 | 3.388855628403834 7 | 134.0 | 0.0 |
| 7 | 55.0 | 0.0 | 572.0 | 1.0 | 35.0 | 0.0 | 231000.0 | 0.8 | 143.0 | 0.0 |
| 8 | 60.0 | 0.0 | 68.0 | 0.0 | 20.0 | 0.0 | 119000.0 | 2.9 | 127.0 | 1.0 |
| 9 | 70.0 | 0.0 | 97.0 | 0.0 | 60.0 | 1.0 | 220000.0 | 0.9 | 138.0 | 1.0 |
| 10 | 53.0 | 1.0 | 1808.0 | 0.0 | 60.0 | 1.0 | 249000.0 | 0.7 | 138.0 | 1.0 |

## B.10 Audit Trace: Fold 10 | Gen: 100%

| # | age | anaemia | creatinine_phosphoki | diabetes | ejection_fractio | high_blood_pressu | platelets | serum_creatinine | serum_sodium | sex |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 62.971623423041535 | 1.0 | 391.0 | 1.0 | 39.0 | 0.0 | 196377.29873843322 | 1.2782974053824918 | 134.0 | 0.0 |
| 2 | 47.817123916091233 | 0.0 | 168.0 | 0.0 | 26.0 | 1.0 | 273271.404895114 | 1.6457190209771928 | 129.0 | 0.0 |
| 3 | 73.568199083844339 | 0.0 | 102.0 | 0.0 | 36.0 | 0.0 | 223578.882083483 65 | 2.390509341705747 5 | 134.0 | 1.0 |
| 4 | 91.952539223499337 | 0.0 | 582.0 | 0.0 | 38.0 | 0.0 | 263358.03 | 1.83 | 134.0 | 0.0 |
| 5 | 80.448687640480133 | 0.0 | 363.0 | 0.0 | 48.0 | 0.0 | 48057.712972399 91 | 1.3282056792639 976 | 135.0 | 0.0 |
| 6 | 81.7353396776918 | 0.0 | 334.0 | 0.0 | 56.0 | 0.0 | 306870.576085948 85 | 1.1891177989256 394 | 132.0 | 1.0 |
| 7 | 60.0 | 0.0 | 897.0 | 1.0 | 45.0 | 0.0 | 297000.0 | 1.0 | 133.0 | 1.0 |
| 8 | 75.0 | 0.0 | 582.0 | 0.0 | 20.0 | 1.0 | 265000.0 | 1.9 | 130.0 | 1.0 |
| 9 | 45.0 | 1.0 | 1876.0 | 1.0 | 35.0 | 0.0 | 226000.0 | 0.9 | 138.0 | 1.0 |
| 10 | 75.0 | 1.0 | 246.0 | 0.0 | 15.0 | 0.0 | 127000.0 | 1.2 | 137.0 | 1.0 |